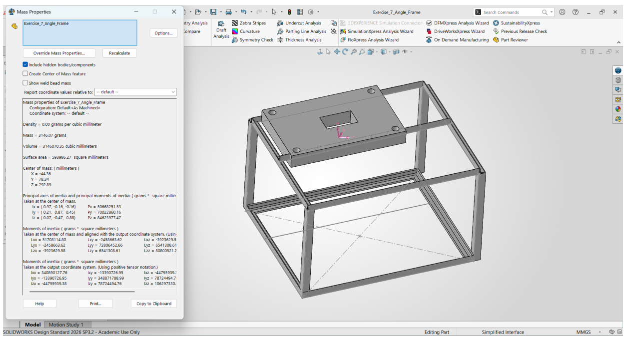
EXERCISE 7

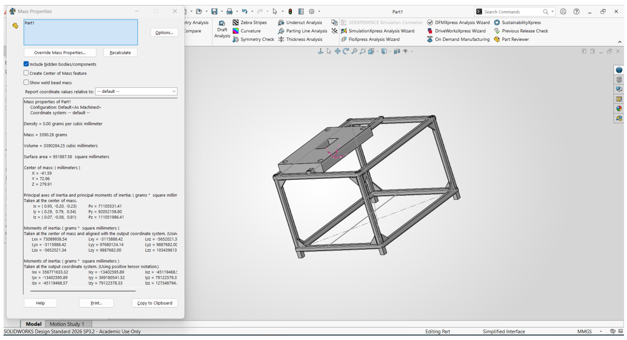
Weldment Design Comparison & Manufacturing Evaluation

Structural Frame – Design A vs Design B

Engineering comparison of structural rigidity, material consumption, fabrication complexity, welding effort, manufacturing cost and assembly.



Design A
Exercise_7_Angle_Frame



Design B
Part1

Prepared as a professional engineering comparison using the two supplied SOLIDWORKS screenshots. Quantitative values are reported only where visible in the Mass Properties panels; other judgments are explicitly identified as engineering observations or inferences.

1. Scope and Source Basis

This report compares the two supplied SOLIDWORKS weldment designs. The comparison is based primarily on the visible frame geometry and the Mass Properties information shown in the screenshots. The report does not assume hidden dimensions, material grade, weld size, load cases, or supplier pricing.

Design A – Exercise_7_Angle_Frame

The screenshot shows a rectangular structural frame supporting a top plate. The frame uses perimeter members and horizontal/intermediate members. The geometry appears relatively open, with fewer visible diagonal braces. The Mass Properties panel reports a mass of **3146.07 g**, volume of **3,146,070.35 mm³**, and surface area of **5,939,836.27 mm²**.

Design B – Part1

The second screenshot shows a more extensively triangulated frame, with visible diagonal members across the side faces in addition to perimeter and cross members. The top plate remains integrated into the structure. The Mass Properties panel reports a mass of **3390.28 g**, volume of **3,390,284.25 mm³**, and surface area of **9,518,887.58 mm²**.

Important data note: both screenshots display density as 0.00 g/mm³. Therefore, material grade and material-cost calculations cannot be established from these screenshots alone. The reported mass and volume are used exactly as displayed, without assigning a material density.

2. Visual Comparison

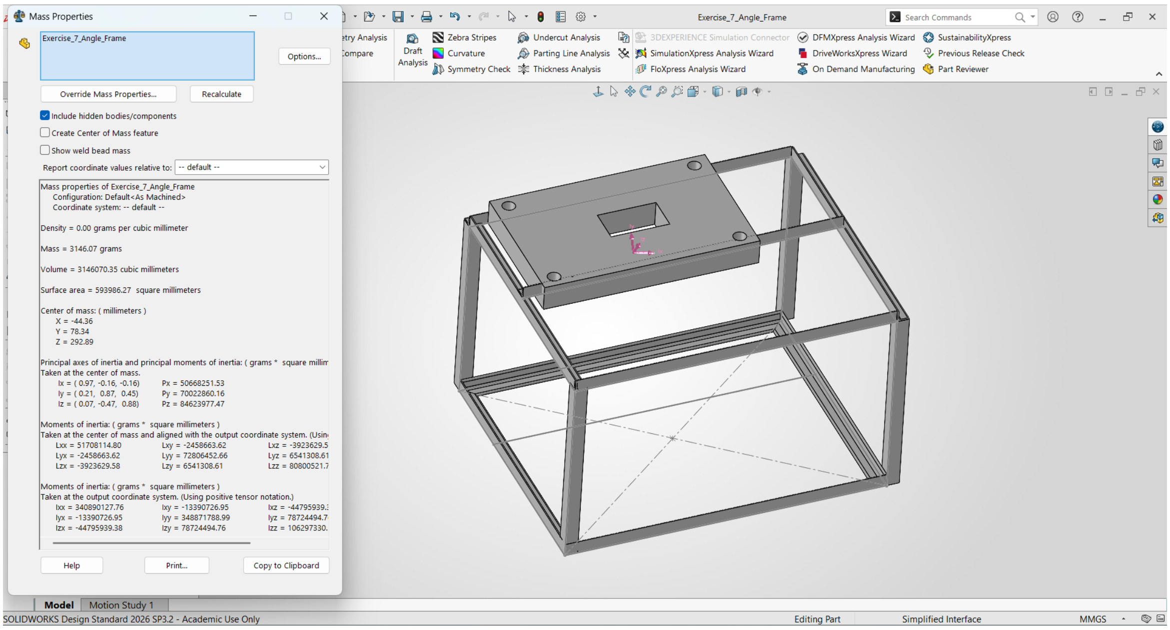


Figure 1 – Design A: Exercise_7_Angle_Frame with Mass Properties.

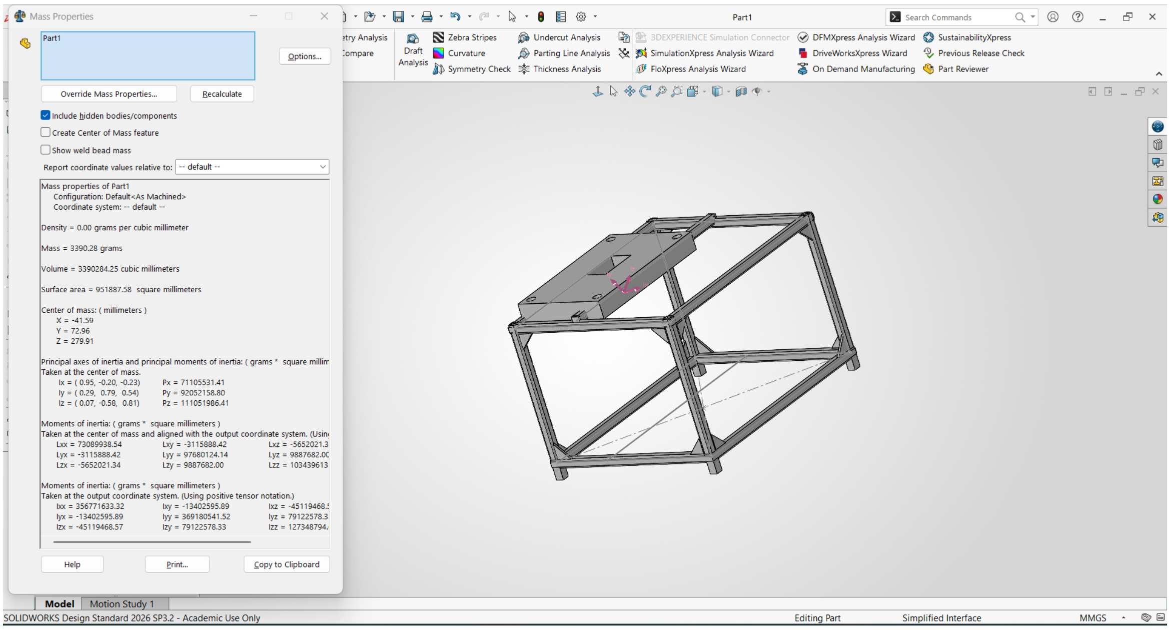


Figure 2 – Design B: Part1 with Mass Properties and additional diagonal bracing.

3. Quantitative Comparison from SOLIDWORKS Screenshots

Metric	Design A	Design B	Change / Interpretation
Reported mass	3146.07 g	3390.28 g	+244.21 g (+7.76%)
Reported volume	3,146,070.35 mm^3	3,390,284.25 mm^3	+244,213.90 mm^3 (+7.76%)
Reported surface area	5,939,836.27 mm^2	9,518,887.58 mm^2	+3,579,051.31 mm^2 (+60.26%)
Material density shown	0.00 g/mm^3	0.00 g/mm^3	Material not quantitatively identified
Visible diagonal bracing	Limited / not prominent	Clearly present	Design B has more triangulation
Visible structural members	Fewer	More	Design B uses more frame members

Mass/material observation: Design B is approximately 7.76% heavier and has approximately 7.76% greater reported volume than Design A. This indicates higher modeled material volume, but it does not by itself establish a higher raw-material purchase cost because material grade, density, stock size and nesting/scrap are not provided.

4. Engineering Design Comparison

Category	Design A	Design B	Assessment
Structural rigidity	Open rectangular frame with fewer visible diagonal members.	Additional diagonal members create triangular load paths.	Design B – likely higher rigidity; confirmation requires FEA/load testing.
Material consumption	Lower reported mass and volume.	Higher reported mass and volume.	Design A – lower modeled material quantity.
Fabrication complexity	Fewer visible members and joints.	More members, intersections and cut/fit locations.	Design A – simpler fabrication.
Welding effort	Fewer visible member intersections.	More visible intersections due to diagonal bracing.	Design A – lower expected welding effort.
Manufacturing cost	Lower material quantity and simpler frame suggest lower cost.	More material and welding suggest higher cost.	Design A likely lower cost if strength requirements are equal.
Ease of assembly	Simpler frame geometry.	More members increase positioning/jigging requirements.	Design A – easier assembly.

Engineering qualification: the rigidity and cost conclusions above are design-level inferences from visible geometry and reported mass/volume. They are not substitutes for structural simulation, weld sizing, fabrication quotations, or physical testing.

5. Manufacturing & Fabrication Analysis

5.1 Fabrication complexity

Design A presents a comparatively straightforward welded frame. Fewer visible members mean fewer individual cut lengths, fewer joints to fit, and fewer weld locations. This generally supports faster fixture setup and simpler inspection.

Design B introduces diagonal members that triangulate the frame. The additional members can improve the structural load path, but they also create additional cut ends, intersections, alignment requirements and welding locations. This is a trade-off between structural performance and production effort.

5.2 Welding effort

For a welded frame, every additional structural member normally creates additional joint preparation, positioning and welding work. Therefore, Design B should be expected to require more welding effort than Design A based on the visible geometry. Actual weld length and weld size are not visible in the supplied screenshots, so no numerical welding time is claimed.

5.3 Trim, fit and cut-list implications

A production-ready weldment benefits from correctly trimmed member intersections. SOLIDWORKS documentation recommends trimming weldment corners so member lengths are calculated accurately. The weldment Cut List can then organize repeated structural members and communicate manufacturing information such as quantities and cut lengths. This is especially relevant to Design B because the additional diagonals increase the number of intersections that must be controlled. ■cite■turn0search0■turn0search3■

5.4 Material and cost

Cost factor	Design A	Design B	Confidence
Modeled material quantity	Lower	Higher	High – based on reported volume/mass
Raw material requirement	Expected lower	Expected higher	Medium – depends on stock/nesting
Cutting operations	Expected fewer	Expected more	Medium – based on visible members
Welding effort	Expected lower	Expected higher	Medium – weld sizes/lengths not supplied
Fixture/jig complexity	Expected lower	Expected higher	Medium
Total manufacturing cost	Likely lower	Likely higher	Low/Medium – no quotations provided

No rupee/dollar cost is estimated because the screenshots do not provide material price, profile purchase length, scrap rate, cutting rate, welding labor rate, or machine time. A real industrial cost comparison should use supplier/production data.

6. Ease of Assembly & Industrial Production Recommendation

Assembly

Design A is easier to assemble because its visible frame is less densely triangulated. Design B requires more members to be positioned and held during welding, increasing the importance of a suitable fixture and controlled assembly sequence.

Structural suitability

Design B is the stronger candidate where frame rigidity and resistance to racking are important, because the diagonal members visibly introduce triangular bracing. However, the screenshots do not contain load cases or FEA results, so the report does not claim a quantified strength increase.

Recommendation: Design B (Part1) is recommended as the more suitable industrial design when structural rigidity is a primary requirement. Its additional bracing provides a more structurally purposeful frame, while the measured increase in mass is only about 7.76% relative to Design A. For a low-load application where minimum cost and minimum fabrication effort dominate, Design A would be preferable.

For production release, Design B should be further optimized by reviewing whether every diagonal is necessary, standardizing member profiles, minimizing unique cut lengths, and using a welding fixture. This could retain the rigidity benefit while reducing manufacturing effort.

7. Final Engineering Evaluation

Criterion	Design A	Design B	Preferred
Structural rigidity	Good	Very Good*	Design B*
Material efficiency	Very Good	Good	Design A
Fabrication simplicity	Very Good	Good	Design A
Welding effort	Very Good	Good	Design A
Ease of assembly	Very Good	Good	Design A
Potential cost efficiency	Very Good	Good	Design A
Industrial structural suitability	Good	Very Good*	Design B*

*Rigidity ratings are qualitative visual assessments. They should be validated with SOLIDWORKS Simulation/FEA or physical testing before a structural safety claim is made.

8. Conclusion

The two designs demonstrate a clear engineering trade-off. Design A is the lighter and simpler frame, with lower reported mass and volume and fewer visible structural members. These characteristics support lower material use, simpler fabrication, lower welding effort and easier assembly.

Design B adds visible diagonal bracing and therefore creates a more triangulated structural system. The additional structure is accompanied by a measured mass increase from 3146.07 g to 3390.28 g, or approximately 7.76%. For an industrial frame in which rigidity and resistance to deformation are important, this additional material can be justified, provided the extra members are actually required by the load case.

Overall recommendation: use **Design B** as the preferred structural-production starting point, then perform a final DFM optimization to remove any non-essential braces and standardize member lengths. Use **Design A** instead when the application has low structural demand and the dominant objective is minimum material and fabrication cost.

9. Data Needed for Final Production Release

Required input	Purpose
Material grade / density	Convert volume into physical mass and validate material cost
Structural load cases	Verify required strength and rigidity
SOLIDWORKS Simulation/FEA	Quantify deformation and stress
Exact profile size and thickness	Validate stock material and cut list
Weld size and weld length	Estimate welding time and cost
Supplier material price	Calculate raw material cost
Cutting and welding cycle times	Calculate manufacturing labor/machine cost
Final Cut List	Define production quantities and cut lengths

SOLIDWORKS weldments are designed around sketch-defined structural members, trimming/extension of members, and manufacturing-oriented cut lists. These capabilities support the workflow used in this exercise and are appropriate for communicating a weldment design to fabrication. ■cite■turn0search7■turn0search10■

Submission note: This report is intended to accompany the Exercise 7 SOLIDWORKS part file and demonstrates the engineering reasoning behind the selected frame design. All numerical values in this report come directly from the two supplied Mass Properties screenshots or are simple calculations from those displayed values.